

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (~1M annotated words across ~10K texts).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Pipeline mechanism, on 1000fps pile-driving footage: object detection to localize the pile, YOLO segmentation and line detection to track painted depth markings, an open-source digit detector to identify line numbers, and audio-visual synchronization to isolate each hammer strike, then **converting pixel-space line displacement into inch/foot measurements** via a depth map.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.
- Field interviews also surfaced that a LIDAR-based measurement approach was cost-prohibitive given the project's thin margins, and that workers frequently crossed the camera's line of sight during recording — **informed the decision to pursue a CV-based approach over LIDAR** for cost-effectiveness.
- Designed proposed capture setup (patent filing in progress): tripod-mounted black-and-white camera at 1000fps on a Jetson Orin GPU, capturing painted chalk markings and LED strips to **measure pile drive-down and rebound height**.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, **Phase 2 LIDAR elimination of markings**.

Projects

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy)

github.com/FavouritePlayer/face-morpher

- Implemented per-triangle affine solve (3-point correspondence → 2×3 matrix), inverse warping with bilinear interpolation, and piecewise Delaunay-triangle warping to produce morphs and mean faces; **all core geometry self-implemented in NumPy**, OpenCV limited to I/O and masking.

Context Custodian — Workspace Hygiene Agent (FastAPI)

github.com/FavouritePlayer/ContextCustodian

- Designed a three-stage prompt-injection detection pipeline — regex prefilter, vector-similarity narrowing, then LLM adjudication — with a **fail-open fallback to regex-only detection** if the LLM call fails, keeping live demos safe.
- Engineered honest-by-construction fix reporting: each fix call returns whether the action **actually executed live or only updated the local audit index**, so the UI never claims a success that didn't happen.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright)

github.com/FavouritePlayer/DealScout

- Built a LangGraph-orchestrated agent that scores resale value with **deterministic arithmetic kept separate from LLM judgment** to avoid hallucinated pricing, then filters and ranks candidates through a structured-output LLM call.

Skills

Languages: Python, SQL

Technologies: PyTorch, Transformers (BART), NLTK, spaCy, NumPy, dlib, scipy, LangGraph, FastAPI, Playwright, Git